



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 29-301-1
06-12-2022

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SPECIFICATION

FOR

METALLIC SURFACE PROTECTION

(HOT-DIP GALVANIZATION / COATING MATERIALS)

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1. SCOPE

This specification covers the minimum technical requirements for manufacturing, testing, and supplying of metallic surface protection material. The protection material is required to improve the surface resistance against corrosion and weathers. The required material shall ensure safe protection of metallic surfaces for new and old electrical network components (such as low voltage poles, medium voltage lattice towers) of Egyptian Electricity Distribution Companies (EDCs). The protection method can be either hot-dip galvanization or coatings applied by airless spray, leading to rapid applications over wide areas.

2. APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

No.	Standard No.	Description
1	ASTM 4541 / ISO 4624 on grit blasted steel / ISO 12944-5	Adhesion test Test Method for Pull-Off Strength of Coatings
2	ASTM B 117-90 / ISO 12944-9	Salt spray test
3	ASTM D2794-93, ASTM D256	Impact test
4	ASTM D522-93a	Bend test These test methods cover the determination of the resistance to cracking (flexibility)
5	ASTM G53	UV resistance test
6	ASTM G8 OR ISO 4628	Corrosion protection / rate
7	ASTM D2370-16	Mechanical Test
8	ASTM A123/EN ISO 1461	Hot-Dip Galvanized Tests



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3. ENVIRONMENTAL CONDITIONS

The chemical and mechanical properties of the protection material should be guaranteed under the environmental conditions given in Table (2). Any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Mean daily temperature	40 °C
Maximum ambient shade temperature	50 °C
Minimum ambient shade temperature	-10 °C
Maximum relative humidity	85 % (option 95%)
Altitude	Up to 1000 m above mean sea-level
Maximum Rainfall	250 mm
Environment	Humid tropical climate with heavily polluted atmosphere

4. TECHNICAL REQUIREMENTS for COATINGS

The Protective coating should be designed to achieve the following requirements:

- The coating materials shall follow the general requirements for the cold galvanizing.
- The coating process should be carried out after the completion of the shearing, punching, assembly, welding and grinding operations. None of these operations or any other mechanical operations should be carried out after completing the coating process.
- Cleaning, sandblasting, degreasing, acid washing and water washing should be carried out before coating.
- To secure lasting adhesion to the subsequent product all surfaces shall be clean, dry and free from any contamination.



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- In case of cold galvanizing by using zinc, the zinc metal shall be of high quality SHG (Special High Grade) with Purity not less than 99.99% according to the standard ASTM A780 and ISO 3549.
- The coating materials shall resist to wear, friction and seawater.
- The coating materials shall withstand at least continuous temperature 120 °C and peak temperature 140 °C.
- Dry film thickness (DFT) per coat layer shall be identified by the supplier and tested by certified lab.
- It could be easily applied on the steel structures established in surface.
- The Pot Life should not be less than 2 hours.
- The test Report shall cover the following:
 - The testing date and the report issuing date.
 - Identification of the specimen tested, source of supply, manufacturer, model and/or serial number, and any other pertinent information.
 - Details of the tested specimen shall include the type of prime paint, the method of paint application, the procedure used to cure it, and the dry film thickness.
 - A statement that the test or tests were conducted in accordance with the methods and procedures as specified herein. If deviations from these methods and procedures were made, they shall be described in the report.
 - When the test is made to check the conformance of the unit specimen to test requirements of a particular specification, the identification or description of the specification shall be included in the report.

5. MARKING

- The following information should be marked on each pot:



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- (a) Manufacturer`s name
- (b) Country of origin
- (c) ASTM standard/ ISO
- (d) Pot life
- (e) Coverage rate
- (f) Manufacturing date: year/month/day
- (g) Expiry date: year/month/day
- (h) Weight in kg/liters
- The marking should be durable and easily legible. Compliance is checked by inspection.

6. TESTING

- All tests shall be carried out by and at the expense of the supplier.
- All tests shall be conducted and /or certified by a nationally recognized, independent testing laboratory accredited in accordance with ISO17025 for the test method referenced in the standard.
- All supplied coatings shall be tested according to the latest ASTM standards at EEHC approved laboratory in the presence of an authorized EEHC representative for the following tests:
 - (a) Heat Resistance
 - (b) Adhesion (tensile shear), pull off
 - (c) Compressive strength
 - (d) Temperature
 - (e) Salt spray test
 - (f) Hardness
 - (g) Corrosion protection



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- (h) Hydrophobic test
- (i) Flexural properties
- (j) Immersion resistance
- (k) Tensile properties

7. GUARANTEE

- The supplier should guarantee the supplied materials against all defects arising out of defective manufacturing for 20 years at least from putting in service.
- The supplier should guarantee that the supplied materials maintain their properties along the shelf life at the specified storing temperature.

8. SUBMITTALS

- The tenderer should submit original catalogues describing the chemical properties of the supplied materials details.
- Copy of recent tests should be submitted with the tender.
- A valid EEHC approval letter must be attached to the offer.
- The supplier should provide all the instructions of use/storing for all surfaces in (Arabic and English).
- The supplier should submit the cure time table of the coating at different time and conditions.
- The supplier should guarantee that all the provided materials are non-toxic and eco-friendly.
- Providing measuring devices (thickness and roughness) accompanied by __ EDC engineer for engineering inspection of the works.

9. TRAINING

- The vendor shall provide a sufficient on-site training to __ EDC workers for free.



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10. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.

TECHNICAL DATA SCHEDULE

No.	ITEMS	Vendor specifications
1	Work life	Hour
2	No of coating layers & thickness	 Layers,mm
3	Painting system	Dry film thickness
	1 st coating	microns
	2 nd coating	microns
	Total dry film thickness (DFT)	microns
4	Country of origin	
5	Manufacturing	
	Year/month/day	
6	Expiry date: year/month/day	
7	Guarantee period	 years
8	Training	Yes / No

I / We guarantee the data given above for the offered material.

Signature:

Date:



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11. SURFACE PREPARATION

SURFACE PREPARATION OF METALS (ACCORDING TO ASTM D6386)

- The Surface abrasion and roughening must achieve two objectives:
 - a) Remove contaminants including oil, grease, rust, scale, old paint etc.
 - b) Increase the surface bonding area and improve mechanical bonding.
- Both objectives are normally achieved at the same time if the correct tool choice is made.

Method one: Surface preparation using Blast Cleaning (For newly-fabricated Components):

Pre-blasting operations:

- All traces of oil and grease contamination shall be removed by a suitable cleaner/degreaser prior to blast cleaning of the surface.
- Where surface defects are revealed during blast cleaning these shall be suitably dressed and the area concerned re-blasted to the original standard.

Blast Cleaning:

- Sand blasting the surface, sand abrasives shall be clean and dry and free from any contaminants, which may adversely affect the performance of the coating.
- The size of the abrasive particles shall give the required surface profile height in accordance with the requirements for the applicable coating system but in any event not less than 75 microns.
- Following blast cleaning all dust/ blast media and other debris shall be removed by sweeping or vacuuming.
- All surfaces shall then be degreased using a suitable cleaner-degreaser, which does not have a residue, such as MEK or Acetone (Non-residue chemical cleaner).
- High build epoxy coatings cured by polyamines or polyamides and cured polyesters



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coatings form strong layer and can up-left the previous applied cured coated layer such as red primer or sacrificial primer. It could only be applied on zinc-rich epoxy.

- The condition of the blasted surface must be maintained until application of the first coat. If this is not possible then the surface shall be re-blasted and degreased prior to application of the coating system.

Method two: Surface preparation using hand tools (For existing Components):

- Roughening the surface using hand tools which do not cause sparks, such as files, wire brushes or emery papers.
- All surfaces shall then be degreased using a suitable cleaner-degreaser, which does not have a residue, such as MEK or Acetone (Non-residue chemical cleaner).

APPLYING PROTECTIVE COATING (EXTERNALLY)

- Apply protective coating (High performance immersion grade coating for long term protection of metallic surfaces under immersion conditions in contact with aqueous solutions. The coating exhibits excellent corrosion resistance) with total thickness 0.3-0.5 mm.



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12. HOT-DIP GALVANIZATION

- تتم عملية الجلفنة بعد اكتمال عمليات القص والتثقيب والتجميع واللحام والتجليخ ولا يجب أن تجرى أى من هذه العمليات أو أي عمليات ميكانيكية أخرى بعد إتمام عملية الجلفنة.
- يتم التنظيف والمراشمة (الترميل) وإزالة الشحوم والغسيل بالأحماض والماء قبل إجراء عملية الجلفنة.
- تجرى عملية الجلفنة بطريقة الغمر على الساخن Hot Dip Galvanized لترسيب طبقة ملساء من الزنك بسمك منتظم يلتصق بشدة بالسطح المراد جلفنته.
- يجب أن يكون معدن الزنك المستخدم من النوع عالي الجودة (SHG) Special High Grade بدرجة نقاوة لا تقل عن 99,99% طبقا للمواصفة القياسية ASTM B6.
- تحتوي أحواض الجلفنة بشكل أساسي على الزنك المنصهر، ولا يزيد مجموع نسب تركيز العناصر الشائبة بالحوض عن 1,5% من وزن الزنك المنصهر طبقا للمواصفة القياسية ISO 1461.
- أقل وزن لطبقة الجلفنة على السطح المعدني (من الداخل والخارج) تكون 610 جم/م² بما يعادل متوسط سمك لا يقل عن 85 ميكرون، أما بالنسبة للمسامير والصواميل فيكون أقل وزن لطبقة الجلفنة 305 جم/م² بما يعادل سمك لا يقل عن 43 ميكرون.

اختبارات الجلفنة

١-١٢ اختبارات التصميم Design Tests

- يجب إجراء الاختبارات التالية بأحد المعامل المعتمدة والمقبولة لدى الشركة القابضة لكهرباء مصر، ويقدم عن كل اختبار شهادة تفيد اجتياز السطح المعدني المجلفن تلك الاختبارات ووفقا للمواصفات القياسية:
 - a- Temperature test (يشترط أن يكون الجهاز معاير)
 - b- Corrosion rate test
 - c- Micro structure test (للتأكد من تكون ٤ طبقات من الجلفنة)

٢-١٢ اختبار نقاوة مادة الزنك بأحواض الجلفنة

- يقدم المورد شهادة اختبار حديثة صادرة عن معمل معتمد موضح بها درجة نقاوة مادة الزنك الموجودة في أحواض الزنك في ذلك الوقت.
- يتم أخذ عينة من مادة الزنك الموجودة بحوض الجلفنة وفي حضور مندوب شركة التوزيع، ليتم عمل الاختبارات اللازمة عليها بأحد المعامل المعتمدة التي تحددها شركة التوزيع لتوضيح درجة نقاوة الزنك طبقا للمواصفة القياسية ASTM B6



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٣-١٢ اختبارات العينة بعد الجلفنة

(أ) الفحص الظاهري

- فحص سمك طبقة الجلفنة ودرجة تماسكها وتجانسها والمظهر الخارجي لها ويجب أن تكون أسطح الأجزاء المجلفنة خالية من أى بقع أو صدأ أو أية ترسيبات إبرية أو هشة.
- يلتزم المورد بتقديم شهادة اختبار حديثة للمسامير والصواميل موضحا بها اجتياز وضع المسامير والصواميل فى حمض لمدة ٤٨ ساعة.

(ب) اختبار الجلفنة

- تجرى اختبارات الجلفنة طبقا للمواصفة القياسية العالمية ISO 1461 للسطح المعدني، والمواصفة القياسية العالمية ISO 10684 للمسامير والصواميل.
- يتم قياس سمك طبقة الجلفنة من خلال أجهزة القياس الدقيقة والمعايرة حديثا وذلك عن طريق أخذ متوسط عدد من القياسات لا تقل عن عشرة قراءات على اجزاء مختلفة من السطح المعدني على أن يكون متوسط القياسات مطابق للقيم الموضحة بالجدول رقم (٣).
- يوزن الجزء المختبر قبل وبعد الجلفنة ويحدد من ذلك وزن الجلفنة لمطابقته بالمطلوب بالجدول رقم (٣).
- اختبار تماسك طبقة الجلفنة مع سطح الجزء المختبر ويتم ذلك بتعريض سطح الجزء المختبر لصدمات ميكانيكية باستخدام مطرقة وزنها ٢٥٠ جم دون ظهور أي علامات شروخ أو تقشير.
- الجدول رقم (٣) يحدد القيم التي يجب ألا تقل عنها نتائج اختبارات الجلفنة.

جدول رقم (٣) اختبار طبقة الجلفنة

نوع الجزء المختبر	الحد الأدنى لمتوسط وزن طبقة الجلفنة جم/م ^٢	الحد الأدنى لمتوسط سمك طبقة الجلفنة (ميكرون)
السطح المعدني (عامود / برج هيكلي)	٦١٠	٨٥
المسامير والصواميل	٣٠٥	٤٣